PRD — In-store Smart Shopping Assistant

1) Overview

Problem statement

In busy malls and retail stores, the physical shopping journey has friction points:

- **Directionless customers:** Shoppers can’t easily find products, offers, or the right aisle/section when staff are busy.
- **Severe checkout delays:** Long queues during peak hours/sales reduce satisfaction and lead to drop-offs.
- **Inventory blindspots:** Customers don’t know real-time stock availability and waste time looking for out-of-stock items.

Proposed solution (product)

Build an **In-store Smart Shopping Assistant** that helps customers:

1. **Navigate** to products/sections (store map + step-by-step guidance).
2. **Check real-time availability** (in-store inventory visibility).
3. **Speed up checkout** via queue visibility + scan-and-go / fast checkout options.

2) Goals, Non-goals

Goals

- Reduce time-to-find-product and time-in-store for common shopping missions.
- Reduce checkout wait time and abandonment during peak periods.
- Increase trust by showing accurate inventory status.

Non-goals (for V1)

- Full omnichannel order management (ship-from-store, returns logistics automation).
- Personalized pricing / loyalty engine rebuild.
- Replacing staff; the product should **support** staff and customers.

3) Users & Use cases

Primary users

- **Shoppers** (walk-in customers)

Secondary users

- **Store staff** (to assist exceptions, inventory, and customer escalations)
- **Store manager/ops** (to monitor adoption, queue load, and issues)

Key use cases

- “Where can I find X?” → guided navigation
- “Do you have size/color Y in stock?” → real-time availability + alternatives
- “I’m in a hurry” → queue time visibility + fastest checkout option
- “I need help” → request assistance / connect to staff

4) As-Is vs Future State

As-Is (current experience)

1. Customer enters store and searches visually or asks staff.
2. Staff might be busy → customer wanders, gets frustrated.
3. Customer picks items but stock uncertainty persists (especially variants).
4. Checkout queue builds → long waits, possible abandonment.

Future state (target experience)

1. Customer opens assistant (QR in-store, store app, or kiosk).
2. Search product → get **aisle/section** + **live stock status**.
3. “Navigate” → map + directions.

4. Cart building → show substitutions if out-of-stock.
5. Checkout → show **queue times** and provide **fast checkout** (scan-and-go / express / self checkout guidance).

5) Scope (MVP)

Customer-facing (MVP)

- Product/Category search (text + barcode scan optional)
- Store map navigation (section-level first; aisle-level where possible)
- Real-time inventory display at store level (available / low stock / out of stock)
- Alternatives & recommendations when out of stock (same category/price range)
- Checkout assistance:
  - Queue time estimates by checkout type (manual, self-checkout)
  - “Fastest checkout” recommendation
  - Scan-and-go (if feasible) OR express checkout guidance

Staff/ops (MVP)

- Dashboard for:
  - Active users
  - Common searches
  - Out-of-stock hits
  - Queue load by counter type
- “Request help” ticket feed (lightweight)

6) Product Requirements (detailed)

6.1 Search & discovery

User story: As a shopper, I want to search for an item and quickly see where it is and whether it’s in stock.

- Must have:

    - Search by product name/category
    - Results show: name, image (if available), price (optional), stock status, location/section
    - “Navigate” CTA
- Edge cases:
    - Multiple similar items → disambiguation
    - No results → show category suggestions

6.2 Navigation / wayfinding

User story: As a shopper, I want step-by-step guidance to the product section.

- Must have:
    - Section-level map with “You are here” (if indoor positioning available) OR “Select your current zone” fallback
    - Directions that are easy to follow (“Go to Floor 1, Aisle 5 – Snacks”)
- MVP fallback:
    - If precise indoor positioning is not available, provide **static map + section highlight**.

6.3 Inventory visibility

User story: As a shopper, I want to know if the product/variant is available before I walk there.

- Must have:
    - Store-level stock for SKU/variant (size/color)
    - Status: Available / Low stock / Out of stock
    - If out of stock: suggest alternatives and/or nearby store (optional)
- Accuracy requirement:
    - Target inventory freshness and reconciliation rules defined with ops.

6.4 Checkout acceleration

User story: As a shopper, I want to minimize time spent in checkout.

- Must have (MVP):

    - Queue-time estimate per checkout type
    - Recommendation: “Fastest option right now”
- Optional (depending on dependencies):
    - Scan-and-go flow (barcode scan → cart → pay)
    - Express checkout QR that routes to a dedicated counter

6.5 Assistance

User story: As a shopper, I want to quickly request help when I can’t find something.

- Must have:
    - “Request help” with category + free text
    - Notify staff dashboard
    - Basic SLA metric (time to first response)

7) Data & integrations

Data sources needed

- Product catalog (names, variants, images, category)
- Store layout map data (sections/aisles)
- Inventory system (stock by SKU/variant at store)
- Checkout counters feed (transactions per minute / queue length / staff input)

Integrations

- POS/checkout telemetry (for queue estimates)
- Inventory API (read)
- Optional payment integration (if scan-and-go)

8) Dependencies

- Inventory system provides reliable store-level stock (freshness + API access).

- Store map data exists (CAD/planogram/section mapping) and is maintained.
- If scan-and-go: payment, fraud, and receipt validation dependencies.
- Store operations alignment: signage (QR placement), staff training, escalation process.

9) Risks & mitigations

- **Inventory accuracy risk:** Incorrect “available” status breaks trust.
  - Mitigation: show “Low stock” band, include disclaimer for last units, frequent sync, exception reporting.
- **Indoor positioning not available:** Navigation could be too generic.
  - Mitigation: start with section-level + clear signage and map highlights; phase in positioning.
- **Checkout dependency complexity:** Scan-and-go may be heavy.
  - Mitigation: ship queue visibility + fastest checkout recommendation first; pilot scan-and-go later.
- **Adoption risk:** Customers may not use the assistant.
  - Mitigation: strong in-store prompts (QR at entry, shelf talkers), clear value proposition, fast loading.
- **Operational load:** Help requests may overwhelm staff.
  - Mitigation: rate limiting, suggested FAQs, prioritize high-value requests.

10) Metrics, KPIs, success measures

North Star metrics

- **Time saved per shopping trip** (proxy: reduced time-to-find and time-in-checkout)

Key KPIs

- Navigation:
  - Median time-to-find-product (baseline vs post-launch)
  - % of searches that start navigation

- Inventory:
  - % sessions viewing stock
  - Out-of-stock encounter rate (searches resulting in OOS)
  - Stock accuracy (system vs cycle-count sampling)
- Checkout:
  - Average checkout wait time (by checkout type)
  - % customers using fastest checkout recommendation
  - Abandonment rate at checkout (proxy signals)
- Adoption:
  - Weekly active shoppers using assistant
  - QR scans / app opens per store per day
- Satisfaction:
  - CSAT/NPS for in-store experience
  - “Was this helpful?” on directions/stock/checkout screens

Success criteria (MVP pilot)

- Reduce median “find product” time by **20%** in pilot stores.
- Reduce average checkout wait time by **15%** during peak hours in pilot.
- Stock accuracy (for top 200 SKUs) ≥ **95%**.
- Weekly active usage: **10%** of store footfall in pilot (or agreed proxy).

11) Rollout plan (high level)

- Phase 1: One store pilot (limited categories + map)
- Phase 2: 3–5 stores, add inventory improvements + queue estimates
- Phase 3: Scale rollout + optional scan-and-go

12) Open questions

- What is the initial platform: web (QR) vs full native app?
- What inventory freshness can we guarantee (e.g., every 5 minutes)?

- What queue measurement method is available (POS telemetry vs manual staff input)?
- What is the definition of “low stock” by category?